Dr. Chao YIN

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Research Interests

3D point cloud processing; **foundation models** (SAM, DINO, CLIP); **LLMs** and **multi-modal 2D/3D LLMs** (LLaVA, LISA); semantic/instance segmentation, weakly-supervised and long-tailed learning; large-scale dataset construction; **remote sensing (RS) and GIS** applications; AI-augmented research workflows.

Research Highlights

- Expertise in **3D computer vision and deep learning**: point cloud segmentation, 3D scene understanding, long-tailed, weakly-supervised, and unsupervised learning for urban, construction, and industrial applications.
- Strong experience in **multi-modal AI systems**: integrating 2D images, 3D LiDAR, and other modality data for automated scene analysis and intelligent decision-making in complex environments.
- **Open-source contributor and practical coder**: Developed and maintained **6+ influential repositories** on GitHub (**140+ stars**) including ResPointNet++ (41 stars), Deep-Learning-On-Point-Clouds (62 stars), SE-PseudoGrid, and SQN implementations; published-paper code released for reproducibility where available.
- Pioneer in **AI-augmented research methodology**: leveraging LLMs (Claude, GPT-5, GLM 4.6) for rapid prototyping, literature synthesis, and research workflow optimization while maintaining rigorous scientific standards.
- **Strong publication and funding track record**: Published **4 first-author Q1 papers** with **1 paper under review/in progress**; PI of **680K RMB** from China Postdoctoral Science Foundation and Guangdong Province Overseas Postdoctoral Talent Support Program.

Education

Hong Kong University of Science and Technology (HKUST) — Hong Kong
Ph.D., Research focus: Point Cloud Processing, Deep Learning, and BIM Modeling — *Jul 2018 – Aug 2022*
Supervisor: Prof. Jack CHENG

Wuhan University — Hubei, China
M.Phil., Major: Cartography and GIS — *Sep 2011 – Jun 2013*
Supervisor: Prof. Zhongliang Fu

Anhui University of Science and Technology — Anhui, China
B.S., Major: Geographic Information System (GIS) — *Sep 2007 – Jun 2011*

Research Experience

Hong Kong University of Science and Technology — Hong Kong
Research Associate — *2026 – Present*

Geography Institute of Guangzhou, Guangdong Academy of Sciences — Guangzhou, China
Associate Researcher — *Sep 2022 – 2026*

- Led research on weakly-supervised 3D point cloud segmentation, long-tailed learning, multi-modal perception, and remote sensing applications.
- Developed large-scale industrial point cloud datasets and benchmarks for MEP scene understanding and safety monitoring.
- Collaborated with Academician Chenghu Zhou on low-altitude economy, UAV-based remote sensing, and intelligent urban scene monitoring projects.

Hong Kong University of Science and Technology — Hong Kong
Research Assistant at BIM Lab — *Aug 2018 – Aug 2022*

- Conducted research on 3D point cloud processing and BIM reconstruction from LiDAR data.
- Developed deep learning methods for automated semantic segmentation of industrial scenes.
- Contributed to Digital Twins for HKUST Campus and scan-to-BIM (ICCBEI 2019 **Best Paper Award**).

Hong Kong Polytechnic University — Hong Kong
Visiting Fellow at Lab of Smart City — *Aug 2017 – Jun 2018*

- Collaborated with Prof. Weizhong SHI on smart city and geospatial data analysis.

Publications

Journal Papers: Under Review / In Progress (1 first-author paper) ** denotes corresponding author*

- **[First Author] C. Yin**, H. Yue, Q. Han, D. Hu, Z. Liang, F. Lin, B. Sun, B. Wang, M. Li, W. Yao, and J. C. P. Cheng, "Industrial3D: A Water-Treatment TLS Point Cloud Dataset and Cross-Paradigm Benchmark for MEP Scene Understanding," arXiv:2603.28660, 2026, *under review*.

Journal Papers: Published (4 first-author Q1 papers; 2 corresponding-author papers: 1 Q1, 1 Q2)

- **[First Author] C. Yin**, B. Wang, V. J. L. Gan, M. Wang, and J. C. P. Cheng*, "Automated semantic segmentation of industrial point clouds using ResPointNet++," *Automation in Construction*, vol. 130, p. 103874, Oct. 2021. (JCR Q1, IF: 10.5)
- **[First Author] C. Yin**, J. C. P. Cheng*, B. Wang, and V. J. L. Gan*, "Automated Classification of Piping Components from 3D LiDAR Point Clouds using SE-PseudoGrid," *Automation in Construction*, vol. 139, p. 104300, Jul. 2022. (JCR Q1, IF: 10.5)
- **[First Author] C. Yin**, B. Yang, J. C. P. Cheng*, V. J. L. Gan*, B. Wang, and J. Yang, "Label-efficient semantic segmentation of large-scale industrial point clouds using weakly supervised learning," *Automation in Construction*, vol. 148, p. 104757, Apr. 2023. (JCR Q1, IF: 10.5)
- **[First Author] C. Yin**, Q. Han, Z. Hou, Y. Liu, A. Dai, H. Hu, J. Yang*, and W. Yao*, "Resolving primitive-sharing ambiguity in long-tailed TLS-based industrial MEP point cloud segmentation via spatial context constraints," *Advanced Engineering Informatics*, vol. 77, p. 105394, Jan. 2027. (JCR Q1, IF: 11.5)
- **[Corresponding Author]** Q. Han, **C. Yin***, Y. Deng, and P. Liu, "Towards Classification of Architectural Styles of Chinese Traditional Settlements Using Deep Learning: A Dataset, a New Framework, and Its Interpretability," *Remote Sensing*, vol. 14, no. 20, Oct. 2022. (JCR Q2, IF: 5.3)
- **[Corresponding Author]** S. Jing and **C. Yin***, "From geometric labels to semantic understanding of indoor building components using multimodal large language models," *Automation in Construction*, vol. 190, p. 107117, Oct. 2026. (JCR Q1, IF: 10.5)
- B. Wang, **C. Yin**, H. Luo, J. C. P. Cheng, and Q. Wang, "Fully automated generation of parametric BIM for MEP scenes based on terrestrial laser scanning data," *Automation in Construction*, vol. 125, p. 103615, May 2021. (JCR Q1, IF: 10.5)
- B. Wang, Q. Wang, J. C. P. Cheng, C. Song, and **C. Yin**, "Vision-assisted BIM reconstruction from 3D LiDAR point clouds for MEP scenes," *Automation in Construction*, vol. 133, p. 103997, Jan. 2022. (JCR Q1, IF: 10.5)
- B. Wang, Q. Wang*, J. C. P. Cheng*, and **C. Yin**, "Object verification based on deep learning point feature comparison for scan-to-BIM," *Automation in Construction*, vol. 142, p. 104515, Oct. 2022. (JCR Q1, IF: 10.5)
- D. Hu, V. J. L. Gan, and **C. Yin**, "Robot-assisted mobile scanning for automated 3D reconstruction and point cloud semantic segmentation of building interiors," *Automation in Construction*, vol. 152, p. 104949, Aug. 2023. (JCR Q1, IF: 10.5)
- B. Wang, Z. Chen, M. Li, Q. Wang, **C. Yin**, and J. C. P. Cheng, "Omni-Scan2BIM: A ready-to-use Scan2BIM approach based on vision foundation models for MEP scenes," *Automation in Construction*, vol. 162, p. 105384, Jun. 2024. (JCR Q1, IF: 10.5)
- Q. Han, Z. Wang, **C. Yin**, Z. Hou, and T. Yao, "Semi-Supervised AI for Architectural Heritage Classification and Style Lineage Discovery in Chinese Traditional Settlements," *ISPRS International Journal of Geo-Information*, vol. 15, no. 5, p. 221, May 2026.
- H. Yue, Q. Wang, **C. Yin**, D. Hu, J. Yang, and Z. Guo, "Deep learning framework for point cloud instance segmentation in construction sites," *Automation in Construction*, vol. 191, p. 107159, Nov. 2026. (JCR Q1, IF: 10.5)

Conference Papers

- **[First Author] C. Yin**, W. Boyu, and C. Jack Chin Pang, "Deep learning-based scan-to-BIM framework for complex MEP scene using laser scanning data," ICCBEI 2019. **Best Paper Award**.
- **[First Author] C. Yin**, J. C. P. Cheng*, B. Wang, and V. J. L. Gan, "Automated Classification of Piping Components from 3D LiDAR Point Clouds using Deep Learning and Squeeze-Excite Mechanism," IPC 2022.

Book Chapters

- **C. Yin**, B. Wang, K. Chen, H. Liu, C. Huang, J. C. P. Cheng*, and Long Chen, "Digital modelling layer: two digital modelling methods," In Q. Lu, X. Xie, A.K. Parlikad, J. Schooling and M. Pitt (editors), *Digital Twins in the Built Environment: Fundamentals, principles and applications*, ICE Publishing, 2022, pp. 101–159.

$ Research Funding & Projects

Guangdong Province Overseas Postdoctoral Talent Support Program (PI, **600K RMB**) Jun 2023 – 2025
- Weakly-supervised semantic segmentation for complex indoor 3D perception using point clouds.

China Postdoctoral Science Foundation (PI, **80K RMB**) Sep 2023 – Dec 2024
- Weakly-supervised semantic segmentation methods for 3D point clouds based on deep long-tail learning.

National Key R&D Program - UAV Cluster Networking for Emergency Rescue 2025
- Proposal writing for key technologies and instant remote sensing applications in urban canyon environments.
- Collaboration with leading institutions on large-scale UAV-based sensing systems.

Xinjiang Talent Development Fund - Low-Altitude Economy Application 2025
- Proposal writing for low-altitude economy project in collaboration with **Academician Chenghu Zhou**.

Space-Air-Ground Integrated Investigation for Land-Based Pollution 2024
- Implementation and reporting for estuarine coastal pollution investigation (Marine Economy Special Project).

National Natural Science Foundation of China (Co-PI, 300K RMB) Jan 2021 – Dec 2023
- Automated identification and extraction of landscape genes for Chinese traditional settlements based on 3D semantic models.

Hong Kong ITF - Automated BIM Generation (3.4 million HKD) Oct 2018 – Dec 2021
- Key participant in automated BIM generation using UAV and indoor 3D laser scanning technologies.

Hong Kong ITF - 3D Geodatabase Framework for Hong Kong (6.06 million HKD) Aug 2017 – Jun 2018
- Key participant in developing this Lightweight 3D Seamless Spatial Data Acquisition System (SSDAS).

Technical Skills & Research Capabilities

3D Vision & Point Clouds: ResPointNet++, PointNet++, KPConv, PTv3; semantic/instance segmentation, detection; weakly-supervised, unsupervised and long-tailed learning

Foundation Models & VLMs: SAM series, DINO series, CLIP, LLaVA; adaptation for segmentation and grounding

LLMs & 3D Multi-modal LLMs: LLM pipelines built from scratch; LLaMA with **LoRA/QLoRA**; prompt engineering, RAG; VLMs (LLaVA, LISA); 3D M-LLMs (Reason3D, Point-LLM, SegPoint, CityAnchor)

Programming & MLOps: Python (expert), PyTorch, TensorFlow, Hugging Face, OpenMMLab, PointCept, Docker

Geospatial & RS: ArcGIS, QGIS, GDAL, CloudCompare, PCL, FME, Revit; remote sensing workflows

Systems: Linux (Ubuntu/CentOS), CUDA, HPC clusters, Git, Vim, Tmux/Screen, Weights & Biases

Academic: LaTeX (expert), paper writing, grant proposals, scientific visualization

Patents

An Automatic Annotation and Pre-training Method and System for Remote Sensing Images, *Invention Patent (China)*, Completed. **Chao YIN**, et al. (First Author)

A Semantic Segmentation Method and System for Remote Sensing Images Based on DINO and Parallel Spatial Feature Fusion, *Invention Patent (China)*, Completed. **Chao YIN**, et al. (First Author)

UAV Base Station Deployment Method, Device, Computer Equipment and Storage Medium, *Invention Patent (China)*, Filed (Mar 2025). Jia Sun, Hongda Hu, Shengnan Zhou, **Chao YIN**, etc. (4th Author)

Multimodal Plant Knowledge Query Methods, Systems, and Computer Equipment, *Invention Patent (China)*, Filed (2023). Zhiwei Hou, **Chao YIN**, et al. (2nd Author)

A Parametric 3D Modeling Method for Buildings, *Invention Patent (China)*, Filed (2019). Yuan Liu, Yunyuan Deng, Wenwu Zheng, **Chao Yin**, etc. (4th Author)

Teaching Experience

Teaching Assistant, HKUST (2020-2021): Building Information Modeling and Smart Construction, Infrastructure Systems Engineering

Lecturer, Hengyang Normal University (2013-2017): OOP, GIS Development, Web GIS, GIS System Design

Academic Service & Professional Activities

Peer Reviewer: Automation in Construction, ISPRS Journal of Photogrammetry and Remote Sensing, Geo-spatial Information Science, Results in Engineering, Remote Sensing

Awards & Honors

Best Paper Award, International Conference on Construction, Building Engineering and Innovation (ICCBEI), 2019

HKUST Postgraduate Scholarship, full doctoral funding (2018-2022)

First Class Prize in Excellent Graduate Scholarship, Wuhan University (2012)

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